Do LLMs Know When They're Making Things Up?

Hallucination detection on HaluEval QA

Susanta Khamrui (26CS91R12)
Project 7, Group 21

November 2026

Abstract

Three ways of spotting a made-up answer are tested on 200 questions from HaluEval QA, using the open model Qwen2.5-3B-Instruct. Each question has one right and one hallucinated answer, so 400 answers are scored. Token probabilities (Task 1) do worse than random guessing. SelfCheckGPT (Task 2) works with its NLI variant (AUC-PR 0.65) and fails with its BERTScore variant (0.32). A logistic regression on all four scores (Task 3) reaches 0.95 on held-out questions. The catch is that answer length alone reaches 0.97: hallucinated answers in HaluEval are about four times longer than right ones. Once length is taken out, the four scores still reach 0.64.

Code. The project, with all code, results and instructions, can be found here:

`https://github.com/susanta75/NLP-Project-2026-Autumn`

To run everything with Docker:

`bash run_docker.sh`	Task 1, 2 and 3 from the saved model outputs
`bash run_docker.sh --recompute`	also reruns the GPU steps of Task 1 and 2

Without Docker: `pip install -r requirements.txt`, then `bash code/run_all.sh`. Each task can also be run on its own (`python code/taskN/taskN.py`). The README lists the data and model sources and where every output is written.

1 Introduction

A language model hallucinates when it gives a fluent, confident answer that is wrong. Nothing in the text itself gives this away, so a detector has to look elsewhere. This project tries three places to look:

- **Task 1:** the model's own token probabilities. Low probability or high uncertainty should mean the model is guessing.
- **Task 2:** SelfCheckGPT [1]. Ask the model the same question many times; if it really knows the answer, the samples should agree with it.
- **Task 3:** combine the scores from Task 1 and 2 in one logistic regression.

The dataset is the QA part of HaluEval [2]. Each question (taken from HotpotQA) comes with a right answer and a hallucinated answer written by ChatGPT. The two kinds of answer look different: right answers are short, like *"no"* or *"Lanoxin"*, while hallucinated answers are usually full sentences. This one fact explains most of what follows.

2 Data, model and metric

Data. 200 of the 10,000 HaluEval QA pairs were sampled with seed 42. Each pair gives two items: the right answer (label 0) and the hallucinated answer (label 1). Only 200 were used because Task 2 needs 20 generated answers per question.

Model. All tasks use `Qwen/Qwen2.5-3B-Instruct` [3] in float16 on one RTX 5060 Ti (16 GB). The prompt is a system line (*"Answer the question concisely and factually."*) followed by the question. The model never sees HaluEval's `knowledge` paragraph, and nothing is fine-tuned.

Metric. Every detector gives each answer a score, higher meaning "more likely hallucinated". Sorting answers by score and walking down the list gives precision P_k and recall R_k at each cut-off k. AUC-PR (average precision) is

$$\text{AUC-PR} = \sum_k (R_k - R_{k-1})\, P_k. \tag{1}$$

Half the answers are hallucinated, so random scoring gives about 0.5 and a perfect detector gives 1. AUROC is reported as well; it is also 0.5 for random.

	Median tokens	Mean tokens	Range	Example
Right answers (label 0)	3.0	3.7	1–13	*no*
Hallucinated answers (label 1)	14.5	16.1	2–46	*Yes, both documentaries...*

Table 1: Answer length in tokens. The hallucinated answer is the longer one in 194 of 200 pairs, and ranking answers by length alone gives AUC-PR 0.972.

Because of this gap (Table 1), answer length is shown as a reference next to every detector.

3 Task 1: Token probabilities

3.1 Method

The prompt x and an answer $a = (a_1, \ldots, a_T)$ go through the model together, in one forward pass. Nothing is generated; the model only reports how likely it found each answer token. Two numbers are taken at every answer position t and averaged:

$$\bar{\ell}(a) = \frac{1}{T}\sum_{t=1}^{T} \log p(a_t \mid x, a_{<t}), \qquad \bar{H}(a) = \frac{1}{T}\sum_{t=1}^{T} \Big(-\sum_v p(v \mid x, a_{<t}) \log p(v \mid x, a_{<t}) \Big). \tag{2}$$

$\bar{\ell}$ is the mean log-probability of the actual tokens, and $\bar{H}$ is the mean entropy of the model's next-token distribution (the inner sum runs over the whole vocabulary). The hallucination scores are $-\bar{\ell}$ and $\bar{H}$. The lowest token log-probability and the highest token entropy were also kept, to see whether averaging hides anything.

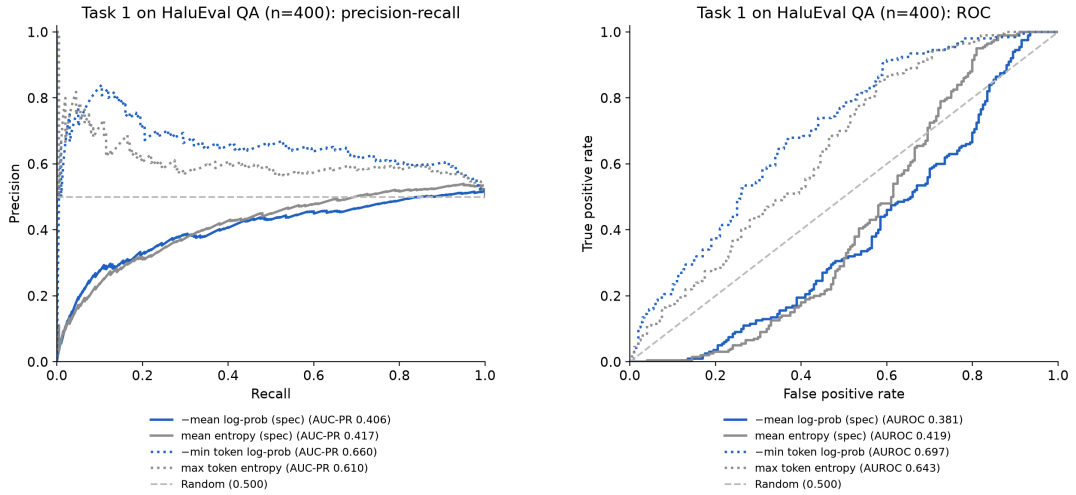
3.2 Results

Both specified scores are worse than random (Table 2, Figure 1). The model is more confident in the hallucinated answer than in the right one in 63% of pairs. The median $\bar{\ell}$ is -3.62 for right answers and -2.49 for hallucinated ones.

One pair shows why. Table 3 lists the token values for both answers.

Score (higher = more suspicious)	AUC-PR	AUROC
−mean log-prob, $-\bar{\ell}$ (specified)	0.406	0.381
mean entropy, $\bar{H}$ (specified)	0.417	0.419
−lowest token log-prob	0.660	0.697
highest token entropy	0.610	0.643
Random	0.500	0.500
Answer length only	0.972	0.971

Table 2: Task 1 on all 400 answers.



(a) Precision–recall **(b)** ROC

Figure 1: Task 1 curves. The two specified scores (solid) are below the random line.

Answer	Token	$\log p$	Entropy	Model's top guess
Right	`no`	−16.30	0.00	`No`
Hallucinated	`Yes`	−15.00	0.00	`No`
	`, both documentaries`	≈ 0	low	same
	`were`	−6.84	1.19	`Tr`
	`directed by`	0.00	0.00	same
	`Johannes`	−15.64	4.84	`Liz`
	`Gren`	−10.52	1.68	`Roberts`
	`zfurthner.`	≈ 0	low	same
Mean over tokens: right / hallucinated		−16.30 / −3.48	0.00 / 0.76	

Table 3: Pair 1290: *"Do the documentaries Traceroute and War/Dance have the same director?"* Right answer *"no"*; hallucinated answer *"Yes, both documentaries were directed by Johannes Grenzfurthner."* Some tokens are grouped to save space.

The model actually knows the answer: its top guess for the first token is *"No"*, and all 20 of its own samples start with "No". But the dataset writes a lowercase *"no"*, which is a different token, so the right answer gets −16.3. The hallucinated answer also has two very unlikely tokens (*Yes* and *Johannes*), but nine easy tokens pull its average up to −3.48. So the mean measures how the answer is written more than whether it is true. Taking the lowest token instead of the mean (0.66) catches cases like *Johannes*, although longer answers also have more chances to contain one rare token.

4 Task 2: Reproducing SelfCheckGPT

4.1 Method

HaluEval has no ready-made samples, so for each question the same model and prompt generated $N = 20$ answers at temperature 1.0 (top-$p = 1$, no top-k, at most 48 new tokens, seed 42 + question id). Both answers of a pair are checked against the same 20 samples.

Scoring uses the authors' own package, `selfcheckgpt` 0.1.7. For an answer r and samples $S^1, \ldots, S^N$, where s_k^n is the k-th sentence of sample S^n:

$$S_{\mathrm{BERT}}(r) = 1 - \frac{1}{N} \sum_{n=1}^{N} \max_k \ \text{BERTScore-F1}(r,\, s_k^n), \tag{3}$$

$$S_{\mathrm{NLI}}(r) = \frac{1}{N} \sum_{n=1}^{N} P(\text{contradiction} \mid r,\, S^n). \tag{4}$$

S_{BERT} is high when the answer shares little wording with the samples. BERTScore [4] uses `roberta-large` [5] with baseline rescaling. S_{NLI} is high when the samples contradict the answer; the NLI model is DeBERTa-v3-large [6] trained on MNLI [7]. No model is trained in this task.

Changes from the brief.

- The brief names `microsoft/deberta-v3-large-mnli`, which does not exist on Hugging Face. `potsawee/deberta-v3-large-mnli` is used instead; it is the model the official code loads.
- `selfcheckgpt` 0.1.7 calls `batch_encode_plus`, which transformers 5 removed. A short patch in `task2.py` adds it back. With it, the package reproduces the example scores in its README to six decimal places.
- BERTScore skips sample sentences of three tokens or fewer. For two questions all 20 samples were single words (e.g. "Canadian"), so those 4 answers get no BERTScore ($n = 396$).

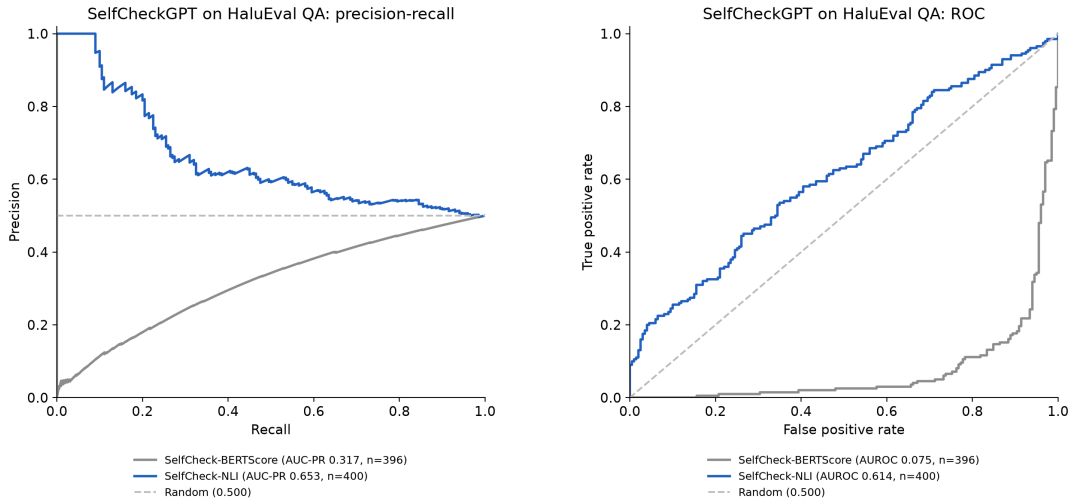
4.2 Results

	AUC-PR	AUROC	Median score right	Median score halluc.	Pairs ranked correctly
SelfCheck-BERTScore ($n = 396$)	0.317	0.075	0.959	0.571	6%
SelfCheck-NLI ($n = 400$)	0.653	0.614	0.920	0.966	67%
Random	0.500	0.500	—	—	50%

Table 4: Task 2 on all answers. "Pairs ranked correctly" is the share of questions where the hallucinated answer gets the higher score.

NLI beats random (Table 4, Figure 2). Its top-ranked answers are mostly right calls: at 20% recall, 83% of the flagged answers really are hallucinated.

BERTScore points the wrong way. It finds the right answer more suspicious in 94% of pairs, and with its sign flipped it would score 0.918. The reason is length again:



(a) Precision–recall **(b)** ROC

Figure 2: Task 2 curves. NLI is above random; BERTScore is far below it.

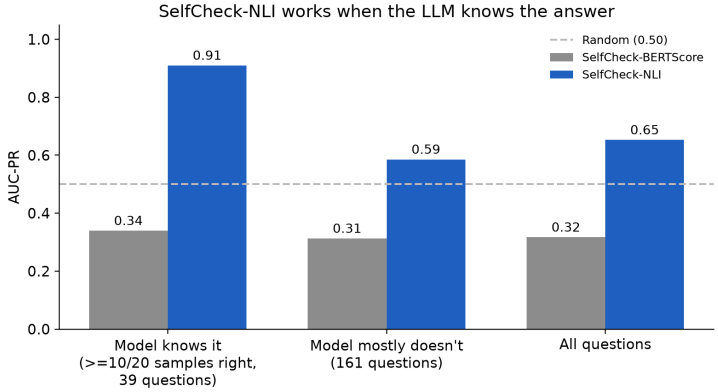


Figure 3: AUC-PR split by whether at least 10 of the 20 samples contain the right answer. With a cut-off of 5 instead, NLI on known questions is 0.89; with 15 it is 0.93.

Pair 117: why BERTScore runs backwards

Question: Jake David Shapiro was best know for being the screenwriter of a film directed by who?
Right: Roger Christian $S_{\text{BERT}} = 1.10$
Hallucinated: Jake David Shapiro was best known for being the screenwriter of a film directed by Steven Spielberg. $S_{\text{BERT}} = 0.34$
Qwen samples: "Jake David Shapiro was best known for being the screenwriter of the film 'Superman II' which was directed by Richard Donner."

Qwen's samples repeat the question. A hallucinated answer that also repeats it shares most of its words with them. A two-word answer shares almost nothing with any full sentence, whatever it says.

4.3 When does self-checking work?

SelfCheckGPT assumes the model usually knows the answer. To test this, the samples containing the right answer were counted for each question. Only 42% of questions have even one such sample, and only 20% (39 questions) have at least 10 of 20.

On the 39 questions the model knows, NLI reaches **0.91**. On the other 161 it drops to 0.59

(Figure 3). Removing the 12 yes/no questions from the 39 still gives 0.90. On questions the model does not know, two things go wrong:

Pair 1169: the model is confidently wrong

Question: Which film director directed the internet video serial "God, Inc.", Roman Karmen or Francis Stokes?
Right: Francis Stokes $S_{\text{NLI}} = 0.999$
Hallucinated: Roman Karmen directed "God, Inc." with Francis Stokes. $S_{\text{NLI}} = 0.965$
Qwen samples: all 20 samples name Roman Karmen

Pair 9763: the model has no idea

Question: In which city did this world games take place when Squash was added as part of the world games?
Right: Lahti, Finland $S_{\text{NLI}} = 0.999$
Hallucinated: The World Games was never held in Lahti. $S_{\text{NLI}} = 0.990$
Qwen samples: "Toronto"; "Riga, Latvia"; "Doha, Qatar"; "... Kuala Lumpur, Malaysia in 2017."

In pair 1169 the samples agree with each other but not with the truth, so the right answer is the one flagged. SelfCheckGPT measures agreement with the model, not with the facts. In pair 9763 the samples are all different, so both answers are contradicted equally.

4.4 Comparison with the paper

The paper reports results on WikiBio, where 73% of sentences are inaccurate, so random already scores 72.96 there. Table 5 therefore also gives the gain over random.

	Paper (WikiBio, NonFact)		This work (HaluEval QA)	
Method	AUC-PR	vs. random	AUC-PR	vs. random
Random	72.96	—	50.0	—
Avg$(-\log p)$	83.21	+10.3	40.6	−9.4
SelfCheck-BERTScore	81.96	+9.0	31.7	−18.3
SelfCheck-NLI	92.50	+19.5	65.3	+15.3
SelfCheck-NLI, known questions	—	—	91.0	+41.0

Table 5: This reproduction against the numbers in the official SelfCheckGPT repository (AUC-PR ×100).

NLI's gain over random (+15) is close to the paper's (+19.5), and NLI beats BERTScore, as in the paper. The other two rows do not reproduce. The likely reasons:

1. **Who wrote the answers.** In WikiBio, the scored text came from the model being tested. In HaluEval, ChatGPT wrote the hallucinations, so Qwen has no special reason to be unsure about them, and they are written in a different style from the right answers.
2. **What the model knows.** The paper used GPT-3 (175B) on biographies; here a 3B model faces multi-hop questions it mostly cannot answer. On questions it does know, NLI does as well as in the paper.
3. **What is compared.** The paper compares sentences with similar sentences. Here one-word answers are compared with full-sentence samples, which hurts BERTScore most.
4. **Size.** 400 answers give wide intervals (about ±0.1 for NLI on the test split).

5 Task 3: Combining the scores

5.1 Method

The 200 *questions* were split 80/20 with seed 42, giving 320 training and 80 test answers. Splitting by question keeps both answers of a question on the same side, so the classifier cannot learn from one answer and be tested on its partner. Each answer is described by its four scores, standardised on the training set, $z = (\bar{\ell}, \bar{H}, S_{\text{BERT}}, S_{\text{NLI}})$. A logistic regression (L2, $C = 1$) learns weights w and a bias b:

$$P(\text{hallucinated} \mid z) = \sigma(w^\top z + b) = \frac{1}{1 + e^{-(w^\top z + b)}}. \tag{5}$$

The four missing BERTScore values are filled with the training median. This is the only model trained in the project, and it takes under a second on a CPU.

Every detector is evaluated on the same 80 test answers. Task 1 and Task 2 appear twice: as raw scores, and as a logistic regression on their own two scores, which is the fairer comparison. The 95% intervals come from 1000 bootstrap resamples of the 40 test questions.

5.2 Results

Detector (held-out 20%, $n = 80$)	AUC-PR	95% CI	AUROC
Task 1: −mean log-prob	0.429	0.40–0.51	0.422
Task 1: mean entropy	0.459	0.42–0.54	0.491
Task 1: LR(log-prob, entropy)	0.602	0.53–0.72	0.676
Task 2: SelfCheck-BERTScore	0.323	0.32–0.34	0.075
Task 2: SelfCheck-NLI	0.651	0.56–0.75	0.602
Task 2: LR(BERTScore, NLI)	0.957	0.91–0.99	0.959
Task 3: LR on all four scores	**0.951**	**0.90–0.98**	**0.951**
Task 3 + lowest log-prob, highest entropy	0.973	0.95–1.00	0.972
Answer length only	0.965	0.94–0.99	0.969
Random	0.500	—	0.500

Table 6: All detectors on the same 80 held-out answers.

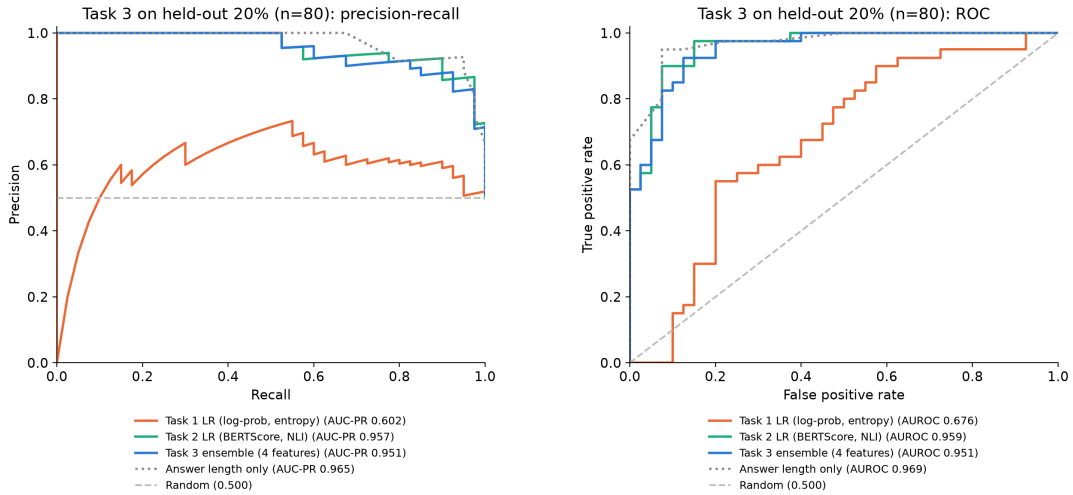
Task 3 reaches 0.951 (Table 6, Figure 4), far above the raw Task 1 and Task 2 scores. It ranks the hallucinated answer higher in 38 of 40 test questions, and labels 87.5% of answers correctly at a 0.5 threshold. But it is level with Task 2's own regression (0.957) and just below length alone (0.965), and the intervals overlap. Over 100 different random splits the picture is the same: Task 1 regression 0.627 ± 0.047, Task 2 regression 0.948 ± 0.025, Task 3 0.948 ± 0.025, length 0.972 ± 0.014.

The learned weights show what happened. BERTScore gets by far the largest weight, -3.08, so the classifier uses it in reverse, exactly as Section 4 suggests. NLI gets $+0.99$, mean log-prob -0.31 and entropy -0.08. Since BERTScore and log answer length are strongly correlated ($r = -0.70$), most of what the regression learns is length.

5.3 Taking length out

Two checks were run, both fitted on the training answers only:

1. **Length as an extra input:** compare a regression on $\log T$ alone with one on $\log T$ plus the four scores. If the scores add nothing, the two should match.



(a) Precision–recall **(b)** ROC

Figure 4: Held-out curves for each task's logistic regression, with answer length for reference.

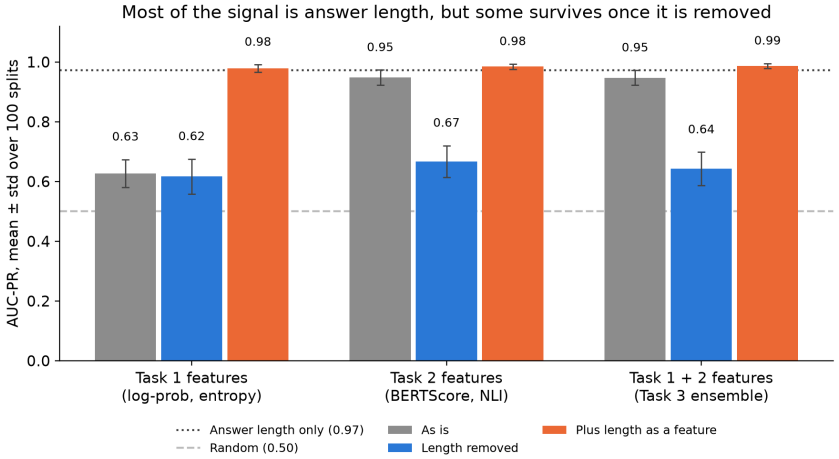


Figure 5: Mean AUC-PR over 100 question-level splits (bars: ±1 sd). Grey: scores as they are. Blue: length removed. Orange: length added as an input.

2. **Length removed:** fit each score f against $\log T$ with a straight line and keep only what the line does not explain,

$$\tilde{f} = f - (\alpha + \beta \log T), \tag{6}$$

then train the regression on these leftovers $\tilde{f}$.

Both were repeated over 100 random splits (seeds 0–99), so the result does not hang on one lucky split.

The scores do carry information beyond length (Figure 5, Table 7). Added to length, they lift AUC-PR from 0.972 to 0.987 and help on 95 of 100 splits. With length removed, 0.64 remains, and three out of four pairs are still ranked correctly.

Mean log-probability changes the most. On its own it was worse than random, but with length removed it ranks 75% of pairs correctly: length had been pushing it the wrong way. NLI moves the other way. After the correction it is no better than random (0.44 on the seed-42 split), so part of its earlier advantage was length too.

Detector (100 splits)	AUC-PR	Pairs correct
LR(length)	0.972 ± 0.014	97%
LR(length + Task 1 + Task 2)	0.987 ± 0.008	98%
LR(Task 1, length removed)	0.616 ± 0.059	74%
LR(Task 2, length removed)	0.667 ± 0.053	71%
LR(Task 1 + Task 2, length removed)	0.642 ± 0.056	75%
−mean log-prob alone, length removed	0.614 ± 0.063	75%

Table 7: Length checks, mean ± sd over 100 splits.

5.4 Where the combined model goes wrong

The model puts the right answer above the hallucinated one in 2 of the 40 test questions. In both, the "hallucinated" answer is in fact correct. In pair 6572, *"... is The Chipmunks"* just rephrases *"Alvin and the Chipmunks"*. Pair 3657 is below; here all 20 Qwen samples say "Canadian", so the model itself is wrong as well.

Pair 3657: a labelling problem in HaluEval

Question: ... songs by Heart, a rock band that first found success in Canada and later in the United States and worldwide, are of which nationality?
Right: American — Task 3 score 0.74
Hallucinated: Heart, the famous rock band, is American. — Task 3 score 0.62

At a 0.5 threshold there are five missed hallucinations. Three are specific but wrong (e.g. "Cancun, Mexico" instead of Daytona Beach), one is nearly a paraphrase of the right answer, and one makes no claim that could be checked:

Pair 2060: an answer that says nothing

Question: Turbidimetric inhibition immunoassay is used in many commercial immunoassays, including this medication sold under which brand name?
Right: Lanoxin — Task 3 score 0.01, $S_{\text{NLI}} = 0.93$
Hallucinated: Turbidimetric inhibition immunoassay is used in many commercial immunoassays, including several medications. — Task 3 score 0.37, $S_{\text{NLI}} = 0.32$

This answer dodges the question instead of inventing a fact, so no sample can contradict it. Self-checking cannot catch an answer that does not say anything.

6 Discussion

What worked. SelfCheck-NLI works when its assumption holds: 0.91 AUC-PR on the questions the model knows, with no training and no access to the model's internals. Splitting by question and always showing a length-only baseline made it clear which numbers mean something.

What did not. The two specified Task 1 scores are worse than random, and SelfCheck-BERTScore is reversed. The Task 3 model looks strong, but mostly because it picks up length through BERTScore.

Why. First, the hallucinations were written by a different model in a different style, so they have little to do with Qwen's own uncertainty, which all three methods rely on. Second, a 3B model answering multi-hop questions without any reference text usually does not know the answer, and then self-checking gives no signal or the wrong one.

Limitations. Only 200 of the 10,000 questions were used, so the intervals are wide. The "known question" split relies on string matching and misses paraphrases. The length correction is a straight line and may not remove all of the effect. A test set with matched lengths was not possible: only 28 pairs have answers within five tokens of each other. A cleaner experiment would score answers that Qwen wrote itself, checked against HotpotQA's gold answers.

7 Conclusion

On HaluEval QA with a 3B open model, token probabilities do not detect hallucinations, and SelfCheckGPT works only on questions the model can answer. The combined regression reaches 0.95 AUC-PR, but nearly all of that comes from answer length. With length taken out, the combined scores rank three out of four pairs correctly, which is a more honest measure of what they can do.

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